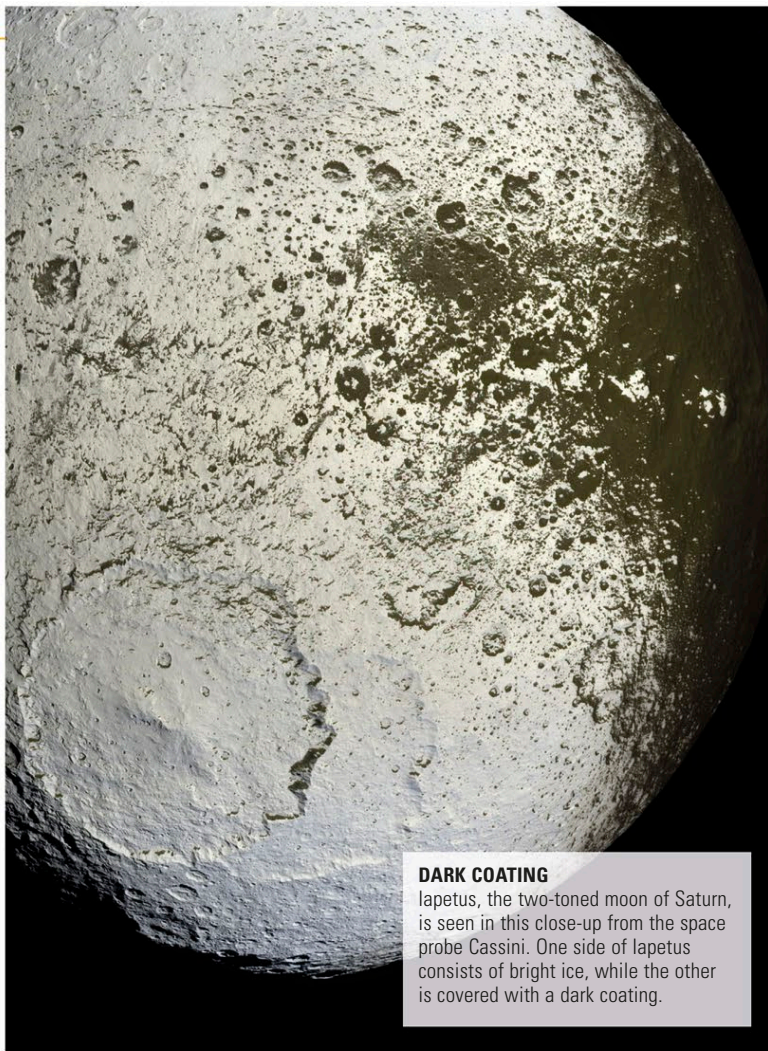




**DARK COATING**

Iapetus, the two-toned moon of Saturn, is seen in this close-up from the space probe Cassini. One side of Iapetus consists of bright ice, while the other is covered with a dark coating.

**MAJOR MOON**

**Iapetus**

**DISTANCE FROM SATURN** 2.21 million miles (3.56 million km)

**ORBITAL PERIOD** 79.33 Earth days

**DIAMETER** 913 miles (1,469 km)

Most of Saturn's inner and major moons orbit in the equatorial plane (also the plane of the rings). Iapetus is an exception, its orbit being inclined by 14.72° to the equatorial plane. Other moons follow orbits with greater inclination, but these are the much smaller, outer moons. Iapetus is Saturn's most distant major moon. It is also in synchronous rotation.

Iapetus was discovered by Italian astronomer Giovanni Cassini while he was working from Paris on October 25, 1671. He noticed that Iapetus has a naturally dark leading hemisphere and a bright trailing hemisphere. The dark region is called Cassini Regio and is covered in material as dark as coal, in contrast to the icy surface



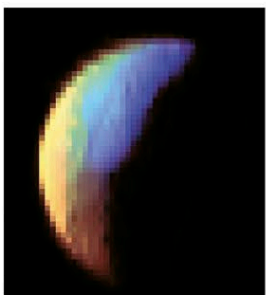
**LANDSLIDE IN CASSINI REGIO**

Land has collapsed down a 9-mile (15-km) high scarp, which marks the edge of a huge impact crater, into a smaller crater. The long distance traveled by the material along the floor indicates that it could be fine-grained.

on the bright side. Although the Cassini probe revealed more of the moon's heavily cratered surface, the origin of the dark material remains a mystery. It has been suggested that the material erupted from the moon's interior or that it is ejecta from impacts on a more distant moon, such as Phoebe (right). A unique feature revealed by Cassini has provided another mystery. It is not known whether a 800-mile (1,300-km) long ridge that coincides almost exactly with the moon's equator is a folded mountain belt or material that erupted through a crack in the surface.

**SURFACE COMPOSITION**

False colors represent Iapetus's vastly different surface compositions. Bright blue signifies an area rich in water ice, dark brown indicates a substance rich in organic material, and the yellow region is composed of a mix of ice and organic chemicals.



**INNER MOON**

**Hyperion**

**DISTANCE FROM SATURN** 919,620 miles (1.48 million km)

**ORBITAL PERIOD** 21.28 Earth days

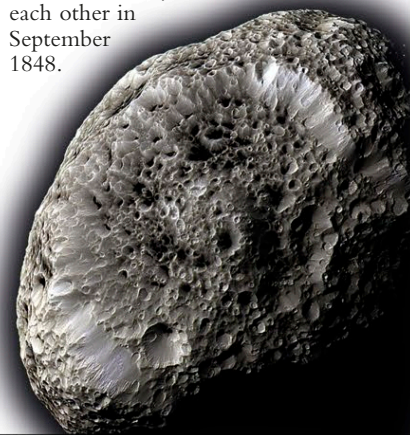
**LENGTH** 224 miles (360 km)

Nothing about Hyperion is typical. First, it is an irregularly shaped moon with an average width of about 174 miles (280 km). This makes it one of the largest nonspherical bodies in the solar system. Second, it follows an elliptical orbit just beyond Saturn's largest moon, Titan (opposite). And, as it orbits, it rotates chaotically: its rotation axis wobbles, and the moon appears to tumble as it travels. Its

**STRANGE CRATERS**

Hyperion has a strange, spongy appearance resulting from its low density and weak gravity.

surface is cratered, and there are segments of cliff faces. Its shape and scarred surface suggest Hyperion could be a fragment of a once-larger object that was broken by a major impact. Even Hyperion's discovery was unusual. Astronomers in the US and England found it independently and within 2 days of each other in September 1848.



**OUTER MOON**

**Phoebe**

**DISTANCE FROM SATURN** 8.05 million miles (12.95 million km)

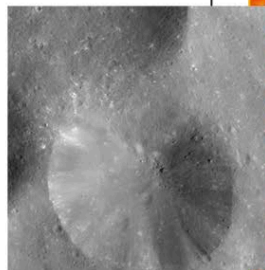
**ORBITAL PERIOD** 550 Earth days

**DIAMETER** 132 miles (213 km)

Phoebe was discovered in 1898 and, until 2000, was thought to be Saturn's only outer moon. Many others are now known to exist. Phoebe has a long orbital period and follows a highly inclined orbit, a characteristic of the outer moons. Phoebe's orbit is inclined by 175.3°, so it travels in a retrograde manner (backward

**CRATER FLOOR**

Debris covers the floor of this impact crater. The streaks inside the crater indicate where loose ejecta has slid down toward the center.



compared to most moons). Half the outer moons orbit this way. Phoebe is by far the largest outer moon; the others are, at most, only 12 miles (20 km) across. From Cassini images, it appears to be an ice-rich body coated with a thin layer of dark material.

**BLASTED PHOEBE**

Phoebe's impact craters were revealed by Cassini in June 2004. They are named after the Argonauts in Greek mythology. The largest, Jason, is about 60 miles (100 km) across.

